5.6 Part – A : Two Marks Questions

Question 122. Define the $'t'$ statistic. **Apr.'11, Apr.'14**

Answer . If $x_1, x_2, ...x_n$ be the small samples (i.e., $n < 30$) drawn from the normal population with mean μ and variance S^2 then the sample mean $\overline{x} = \frac{\sum_{i=1}^{n} x_i}{n}$ and standard deviation $S^2 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n-1}$ then the statistic is

$$t = \frac{\overline{x} - \mu}{S/\sqrt{n}} \text{ (or) } \frac{\overline{x} - \mu}{s/\sqrt{n-1}}.$$

Question 123. Write down the 95% and 99% of confidence or fiducial limits of μ. **(OR)** Give the 95% and 99% confidence interval of the population mean interms of mean and SD of small sample **Nov.'15**

Answer . 1. 95% confidence limits of mean $\mu = \overline{x} \pm t_{0.05} \frac{S}{\sqrt{n}}$.
2. 99% confidence limits of mean $\mu = \overline{x} \pm t_{0.01} \frac{S}{\sqrt{n}}$.

Question 124. State the properties of t-distribution. **Nov.'10, Apr.'13**

Answer . 1. The pdf of t distribution is $f(t) = \left(1 + \frac{t^2}{\nu}\right)^{-\frac{\nu+1}{2}}$, where $t = \frac{\overline{x}-\mu}{S/\sqrt{n}} = \frac{\overline{x}-\mu}{s/\sqrt{n-2}}$, $\int_{-\infty}^{\infty} f(t)\, dt = 1$.

2. The values of t varies from $-\infty$ to ∞.
3. It has symmetrical about $t = 0$ and mean zero.
4. The variance of the $t-$distribution is greater than 1 but tends to 1 as $n \to \infty$. As $\nu \to \infty$ then $t-$distribution becomes normal.
5. The variance $= \frac{\nu}{\nu-2}$, if $\nu > 2$ and $\mu_2 > 1$ always.

Question 125. State some applications (or) uses of $t-$distribution. **Apr.'15**

Answer . 1. To test the significance of a single mean.
2. To test the significance difference between the two sample means.

3. To test the significance of the correlation coefficient.
4. To test the significance of observed partial and multiple correlation coefficients.

Question 126. State the assumptions of t–distribution. **Apr.'11**

Answer . 1. The parent population from which the sample is drawn is normal.

2. The sample observations are independent, i.e., the sample is random.
3. The population standard deviation is unknown.
4. Sample size $n < 30$.

Question 127. State the properties of χ^2 distribution. **Nov.'14**

Answer . 1. The sum of independent chi-square variables is also chi-square variable.

2. Chi-square distribution approaches to normal distribution when $n \to \infty$.
3. The Mgf of chi-square distribution is $(1-2t)^{-\nu/2}$.
4. If k is the number of linearly independent constraints then $\nu = n - k$.
5. For $p \times q$ contigency table, the degrees of freedom is $(p-1)(q-1)$.

Question 128. State some applications or Uses of χ^2 distribution **Nov.'10, Nov.'12**

Answer . 1. To test the 'goodness of fit'.

2. To test the 'independence of attributes'.
3. To test the homogeneity of independent estimates of the population variance.
4. To test the hypothetical value of the population variance is σ^2.

Question 129. What the conditions for application of chi-square test. **(OR)** State the conditions under which chi-square test of goodness fit is valid. **Nov.'11, Apr.'12, Apr.'15, Nov.'15**

Answer . This is an appropriate test for large values of n. For the validity of chi-squáre test of 'goodness of fit' between the theory and experiment, the following conditions must be satisfied.

1. The sample observations should be independent.
2. Constraints on cell frequencies must be linear.
3. N, the total frequency, should be reasonably large say > 50.
4. No theoretical frequencies is less than 5. If it is 5, then the application of χ^2 test, it is pooled with the succeeding or preceding so that the pooled frequency is more than 5.

Question 130. A random sample of size 16 has 53 as mean. The sum of squares of the deviations taken from mean is 135. Can this sample be regarded as taken from the populations having 56 as mean?

Solution . Step 1: Given $n = 16$, $\overline{x} = 53$, $\mu = 56$ and $\sum(x-\overline{x})^2 = 135$.

$$\therefore \quad s = \sqrt{\frac{\sum(x-\overline{x})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3.$$

Step 2 : Null hypothesis $H_0 : \ \mu = 56$.

Alternative hypothesis $H_1 : \mu \neq 56$.

Step 3: Calculated value of z

$$\text{Test statistic } z = \frac{\overline{x}-\mu}{\sigma/\sqrt{n}} = \frac{53-56}{3/\sqrt{16}} = 4$$

Step 4: Tabulated value of z at 5% level of significance $= 2.58$

Step 5: Conclusion :

At 5% level, the calculated value of $z <$ tabulated value of z.
$\therefore$ Accept H_0. There is no significance difference of the population mean and sample mean.

Question 131. Write the uses of $F-$ distribution.

Nov.'13, Nov.'14, Nov.'15

Answer . 1. It is used to test whether two independent samples have been drawn from the normal populations with same variance σ^2

2. It is used to test whether the two independent estimates of the population variance are homogeneous or not.

Question 132. What are the applications of $F-$ test. **Apr.'12**

Answer . 1. It is used to test whether two independent samples have been drawn from the normal populations with same variance σ^2.

2. It is used to test whether the two independent estimates of the population variance are homogeneous or not.

Question 133. Define a 'Attributes'. **Nov.'13**

Answer . A mathematical process used to analyze the characteristics of a given population of subjects. Attributes have only two possible ratings (negative or positive) expressed as acceptable or unacceptable, desirable or undesirable, good or bad, etc.

Question 134. A random sample of size 16 has 53 as mean. The sum of squares of the deviation from mean is 135. Can this sample be regarded as taken from the population having 56 as mean? **Nov.'12**

Solution . Given $n = 16$, $\bar{x} = 53$, $\mu = 56$ and $\sum(x-\bar{x})^2 = 135$.

$\therefore s = \sqrt{\frac{\sum(x-\bar{x})^2}{n-1}} = \sqrt{\frac{135}{15}} = 3.$

Null hypothesis $H_0 : \mu = 56$. Alternative hypothesis $H_1 : \mu \neq 56$.

$$\text{Test statistic } z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}} = \frac{53-56}{3/\sqrt{16}}$$

$$\therefore z = 4$$

Tabulated value of z at 5% level of significance $= 2.58$
At 5% level, the calculated value of $z <$ tabulated value of z.
$\therefore$ Accept H_0. There is no significance difference of the population mean and sample mean.

Question 135. Write the probability density function of $F-$ test.
Apr.'13

Answer . The probability distribution of F is given by

$$f(F) = k(F)^{\frac{\nu_1}{2}-1}\left[1+\frac{\nu_1}{\nu_2}\right]^{-(\nu_1+\nu_2)/2}, \quad 0 \leq F < \infty.$$

Question 136. Define Student's $t-$ test for difference of means of two samples.
Nov.'11

Answer . Consider two independent small samples of sizes n_1 and n_2. Let $\overline{x}_1$ and $\overline{x}_2$ be the sample means and s_1^2 and s_2^2 be the sample variances. To test the difference between the means are significant by using test statistic

$$t = \frac{\overline{x}_1 - \overline{x}_2}{S\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}. \quad \text{Where } S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1+n_2-2}$$

The degrees of freedom of this statistic is $\nu = n_1 + n_2 - 2$.

Question 137. Define standard error. **Apr.'11, Apr.'15**

Answer . The standard deviation of the sampling distribution of a statistic is known as standard error.

Question 138. Define chi-square test for goodness of fit.
Apr.'11, Apr.'12

Answer . If O_i and E_i $(i = 1, 2, ...n)$ be the set of observed (experimental) and expected (theoretical) frequencies then χ^2 is defined as

$$\chi^2 = \sum_{n=1}^{n}\left[\frac{(O_i - E_i)^2}{E_i}\right]$$

Question 139. What do you mean by degrees of freedom **Apr.'15**

Answer . The number of degrees of freedom (ν) is the total number of observations(n) less than the number of independent constraints(k) imposed on the observations. Thus $\nu = n - k$.

Question 140. Write down the chi-square statistic for the contingency table.

a	b
c	d

Apr.'14

Answer . The value of χ^2 test for 2×2 contingency table can be calculated as follows:

Table of Observed Frequencies and Expected frequencies

	A_1	A_2	Total
B_1	a	b	$a+b$
B_2	c	d	$c+d$
Total	$a+c$	$b+d$	N

	A_1	A_2
B_1	$\frac{(a+b)(a+c)}{N}$	$\frac{(a+b)(b+d)}{N}$
B_2	$\frac{(a+c)(c+d)}{N}$	$\frac{(b+d)(c+d)}{N}$

Where $N = a+b+c+d$.

Then the χ^2 statistic is

$$\chi^2 = \frac{N(ad-bc)^2}{(a+b)(c+d)(a+c)(b+d)}.$$

Question 141. Write the assumptions of $F-$test.

Answer . 1. The populations for each sample must be normally distributed.

2. The samples must be random also independent.

3. The ratio of the variances must be greater than 1. So keep always larger variance of the sample in the numerator.